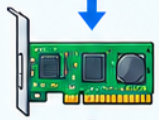
ESXi NETWORKING – COMPLETE VISUAL GUIDE (PHASE 1)

From Physical Networking to VMware Virtual Networking (End to End Flow)

1 Physical Networking (Traditional)

In a physical environment, a computer (PC) connects to the network using a physical NIC, which connects to a switch and then to a router.

PC
(Operating System)



NIC
(Network Interface Card)
Physical network adapter

Switch
(Connects multiple devices)

Router
(Routes traffic to other networks / Internet)

Network / Internet

2 VMware/ESXi Networking (Virtual)

In a VMware/ESXi environment, a VM uses a vNIC, which connects to a Port Group. The Port Group is part of a vSwitch. The vSwitch maps to a physical NIC (vmnic) on the ESXi host, which then connects to the physical switch and network.

Virtual Machine (VM)
Each VM has a **vNIC** (virtual network adapter)

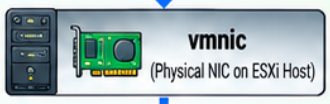
vNIC
Virtual Network Interface Card (inside the VM)

Port Group
(Defines network settings like VLAN, security, etc.)

Port Group
Collection of vNICs with same network settings

vSwitch
(Virtual Switch)

vSwitch
Virtual network switch inside ESXi



vmnic
(Physical NIC on ESXi Host)

vmnic
Physical Network Interface on ESXi host

Physical Switch
(Network Switch)

Physical Switch
Connects ESXi host to physical network

Physical Network / LAN / Internet

3 Mapping: Virtual to Physical

VMware/ESXi (Virtual)		Physical Networking
VM	⟷	PC / Server
vNIC	⟷	NIC
Port Group	⟷	Switch Port / VLAN
vSwitch	⟷	Switch
vmnic	⟷	Physical NIC
Physical Switch	⟷	Physical Switch
Physical Network	⟷	Network / Internet

4 Key Components in ESXi Networking

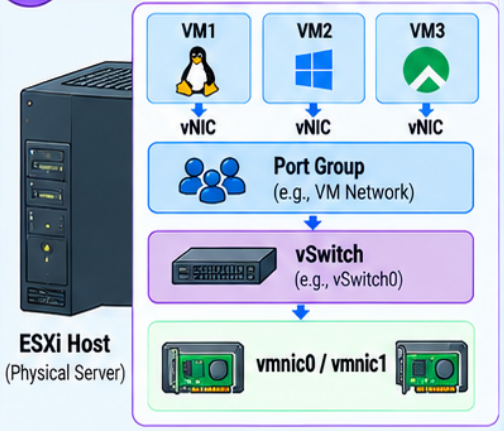
vNIC	Virtual network adapter inside a VM.
Port Group	Groups vNICs with same network configuration (e.g., VLAN, security).
vSwitch	Virtual switch inside ESXi that connects port groups and maps to physical NICs.
vmnic	Physical NIC on ESXi host (connects to physical switch).
Physical Switch	Network switch in the physical environment.

5 Important Points (Phase 1)

- VMs use vNIC, not physical NIC.
- vNIC connects to a Port Group.
- Port Group is part of a vSwitch.
- vSwitch maps to a vmnic (physical NIC).
- vmnic connects to the physical switch.
- Correct network configuration (IP, VLAN, etc.) is important.
- This is the basic networking flow we need to understand in Phase 1.

6 End to End Example (Real Scenario)

7 Inside ESXi Host – Simple View



8 Common Port Groups (Examples)

Port Group Name	Purpose
VM Network	General VM traffic (default)
Management Network	ESXi management traffic (Host IP)
vMotion	Live migration traffic
Storage	iSCSI / NFS storage traffic
Custom VLAN	Specific network with VLAN ID

HYPERVISOR Types, Architecture and Comparison

How a Hypervisor Manages Physical Resources to Run Multiple Virtual Machines

1 What is a Hypervisor?

A **Hypervisor** is a software/platform that manages and allocates the physical hardware resources (CPU, RAM, storage, network) to multiple virtual machines (VMs).

Physical Server (Hardware)

CPU | RAM | Storage | NIC

HYPERVISOR
(e.g., VMware ESXi)

- **VM1** Linux
- **VM2** Windows
- **VM3** Rocky Linux

2 How Hypervisor Allocates Resources

The hypervisor divides and allocates the physical server resources among multiple VMs as per configuration.

Physical Server Resources

CPU	RAM	Storage	Network
16 Cores	64 GB	2 TB	2 NICs

HYPERVISOR
(Resource Management and Allocation)

VM1 Linux	VM2 Windows	VM3 Rocky Linux
2 vCPU	4 vCPU	2 vCPU
8 GB RAM	16 GB RAM	8 GB RAM
100 GB Disk	200 GB Disk	100 GB Disk
1 vNIC	1 vNIC	1 vNIC

3 Types of Hypervisor

Type 1 Hypervisor (Bare-Metal)

- Runs directly on physical hardware.
- No general-purpose operating system required.
- Designed for servers and data centers.
- Better performance, stability and security.

Physical Hardware
(CPU, RAM, Storage, NIC)

Type 1 Hypervisor
(e.g., VMware ESXi)

- **VM1** Linux
- **VM2** Windows
- **VM3** Rocky Linux

Type 2 Hypervisor (Hosted)

- Runs on top of an existing operating system.
- Installs as an application in the host OS.
- Commonly used for desktops and labs.
- Easier to install and use.

Physical PC
(Host Hardware)

Host Operating System
(e.g., Windows)

Type 2 Hypervisor
(e.g., VMware Workstation)

- **VM1** Linux
- **VM2** Windows
- **VM3** Rocky Linux

4 Type 1 vs Type 2 Hypervisor – Comparison

Feature	Type 1 (Bare-Metal)	Type 2 (Hosted)
Deployment	Runs directly on physical hardware	Runs on top of an existing OS
Host OS required?	No	Yes (e.g., Windows, Linux, macOS)
Example	VMware ESXi, Microsoft Hyper-V (Citrix XenServer)	VMware Workstation, Oracle VirtualBox
Common use	Servers, data centers, production	Desktops, labs, learning, testing
Performance	Higher (more direct hardware access)	Lower (depends on host OS)
Stability	More stable	Depends on host OS
Security	Generally more secure	Depends on host OS security
Ease of installation	Requires server installation	Easy (install as an application)
Use cases	Production, enterprise, cloud	Development, testing, training

5 Real-world Examples

Type 1 Hypervisors

- VMware ESXi
- Microsoft Hyper-V
- citrix XenServer

Type 2 Hypervisors

- VMware Workstation
- Oracle VirtualBox
- Parallels Desktop

6 Key Takeaways

- Hypervisor manages and allocates physical resources to multiple VMs.
- Type 1 runs directly on hardware (e.g., VMware ESXi).
- Type 2 runs on top of an existing OS (e.g., VMware Workstation).
- Type 1 is commonly used in servers and data centers.
- Type 2 is commonly used for desktops and labs.
- Both allow you to run multiple operating systems on one machine.

VMWARE & ESXi – COMPLETE VISUAL GUIDE

Introduction, ESX vs ESXi, ESXi vs Hyper-V, Installation, Custom ISO, DCUI And Boot Process

8 VMware Introduction

vmware → Now part of BROADCOM

VMware has been a leading vendor in the virtualization industry and is now part of Broadcom.

Important VMware Technologies / Products

- **ESXi** — Bare-metal hypervisor (Type 1)
- **vCenter / vSphere** — Centralized management
- **vSAN** — Software-defined storage
- **NSX** — Network virtualization and security

Our current learning focus is VMware ESXi.

9 ESX vs ESXi

ESX (Older Architecture)	ESXi (Newer Architecture)
• VMware ka older hypervisor architecture tha. • Ismein traditional Service Console included thi. • Service Console ek Linux-based management environment tha.	• Purpose-built architecture. • Traditional ESX Service Console nahi hoti. • Smaller, more secure and efficient. • Designed only for running and managing VMs.

ESX Architecture

Service Console (Linux-based) | Management Components

VMkernel (Core Hypervisor)

↓

Physical Hardware (CPU, RAM, Storage, NIC)

ESXi Architecture

Management Components (e.g., DCUI, ESXi Shell, Console)

VMkernel (Core Hypervisor)

↓

Physical Hardware (CPU, RAM, Storage, NIC)

Remember: ESX = old | ESXi = modern VMware bare-metal hypervisor family

10 ESXi vs Hyper-V

VMware / Broadcom	Microsoft
ESXi	Hyper-V
Type 1 Hypervisor (Bare-Metal)	Type 1 Hypervisor (Bare-Metal)

Feature	VMware ESXi	Microsoft Hyper-V
Vendor	VMware / Broadcom	Microsoft
Type	Type 1 (Bare-Metal)	Type 1 (Bare-Metal)
Architecture	ESXi (VMkernel)	Hyper-V (Windows hypervisor)
Management	vSphere Client / vCenter	Hyper-V Manager / Windows Admin Center
Ecosystem	VMware ecosystem (vSphere, vSAN, NSX)	Microsoft ecosystem (Windows Server, SCVMM)
Features	Advanced enterprise features	Integrated with Windows environment
Licensing	Subscription based (Broadcom)	Included with Windows Server
Common Use	Enterprise, data centers	Enterprise, organizations using Microsoft stack

Both ESXi and Hyper-V provide virtualization, but they differ in architecture, management tools, features, licensing and ecosystem.

11 ESXi Installation

General Installation Flow (Physical Server)

ESXi ISO → Bootable Media (USB / DVD) → Server Boot (from media) → ESXi Installer → Select Target Disk → Install ESXi → Reboot → ESXi Host Ready

Nested Lab Installation (Using VMware Workstation)

ESXi ISO → VMware Workstation VM → Boot from ISO → Install ESXi → ESXi VM (Running inside Workstation)

12 Customized ESXi ISO

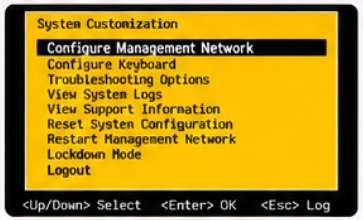
Sometimes standard ESXi image may not include required vendor components/drivers for specific physical hardware.

- **Standard ESXi Image** (from VMware)
- \+ **Required Components** (Drivers, Hardware Add-ons)
- → **Customized ESXi Image** (with required components)

Key Point:
ESXi installation mein hardware **compatibility** important hai.

13 DCUI (Direct Console User Interface)

DCUI is a local, text-based interface available on the ESXi host console for basic configuration and troubleshooting.



Common Tasks in DCUI

- Configure Management Network
- IP Address
- Subnet Mask
- Default Gateway
- DNS
- Hostname
- Restart Management Network
- Troubleshooting Options

Easy Formula:
DCUI → Local/basic ESXi configuration

14 ESXi Boot Process

ESXi boot process loads the necessary components and services to bring the host into a ready state.

Power ON → BIOS / UEFI (Initialize hardware) → Boot Device (USB / HDD / DVD) → ESXi Bootloader (Loads ESXi) → VMkernel (Core component) → Drivers / Modules (Load hardware drivers) → Services (Start system services) → ESXi Host Ready

VMkernel

VMkernel is the core component of the ESXi architecture.

It manages hardware resources, runs VMs and provides infrastructure services.

We will cover VMkernel in more detail in a later phase.

Key Takeaways

- ESXi is a bare-metal hypervisor.
- Boot process loads VMkernel, drivers and services.
- Host becomes ready to run and manage VMs.

VMWARE ESXi – MANAGEMENT AND OPERATIONS

Host Client, vSphere Client, Connectivity, Licensing, Services and Upgrade

15 ESXi Host Client / vSphere Client

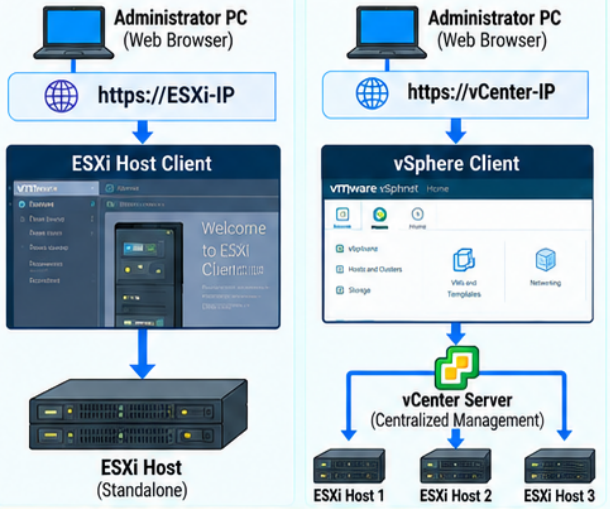
Two Ways to Manage ESXi

1. ESXi Host Client (Standalone)

- Used to manage a single ESXi host directly.
- Accessed via web browser.
- No vCenter required.
- Suitable for labs, small environments or basic management.

2. vSphere Client (With vCenter)

- Used to manage multiple ESXi hosts through vCenter.
- Accessed via web browser.
- Provides centralized management.
- Used in production environments.

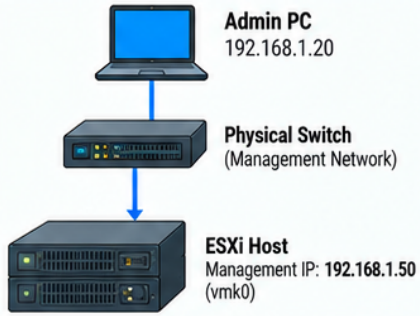


ESXi Host Client and vCenter-based vSphere Client will be covered in more detail in a future phase.

16 Connecting to ESXi

ESXi is given a management network IP address. If network configuration and connectivity are correct, you can manage the ESXi host using a web browser.

Network Diagram



Steps to Connect

1. Configure management network on ESXi (DCUI or during installation).
2. Set IP address, subnet mask, default gateway and DNS.
3. Ensure network connectivity (same network/ routing).
4. Open web browser and visit: https://<ESXi-IP>
5. Login with ESXi credentials.

17 ESXi Licensing

- VMware environment mein available features and permitted usage licensing/subscription par depend kar sakte hain.
- Different editions/licensing levels mein different features available ho sakte hain.
- VMware ab Broadcom ka hissa hai, aur licensing time ke saath change hui hai.
- Current licensing, edition selection ya certification decision ke liye latest official Broadcom/VMware information verify karni chahiye.

Common ESXi Licensing Options (Example)

Edition (Example)	Typical Use
Free (Limited)	Basic lab/learning (limited features)
Essentials	Small environments
Standard	Production environments (core features)
Enterprise / Enterprise Plus	Advanced features (vMotion, HA, vSAN, etc.)

Note: Licensing options, names and features may change over time. Always check the latest official information.

18 ESXi Services

ESXi includes different management and system services and components.

- **Management components** (e.g., Host Client, management agents)
- **SSH** (Remote access, usually disabled by default)
- **Time synchronization** (e.g., NTP)
- **Networking** (vSwitches, vmkernel, management)
- **VM-related services** (Management of virtual machines)

Security Rule: Har service ko bina requirement enable nahi karna chahiye. Unnecessary services disable rakhna chahiye.

Common ESXi Services (Examples)

Service / Component	Purpose
Management Agent	Host management and communication.
SSH	Remote shell access (enable only when needed).
NTP	Time synchronization.
ESXi Shell	Local command-line interface (usually disabled).
SNMP	Monitoring and management (in specific environments).
Syslog	Send logs to remote server.
vCenter Agent	Communication with vCenter (when managed by vCenter).
Network Services	vSwitch, vmkernel, management networking.
VM Services	Run and manage virtual machines.

19 ESXi Upgrade

Production environment mein upgrade se pehle proper planning aur compatibility checking important hoti hai.

Key Considerations

- Check hardware compatibility (HCL).
- Review release notes.
- Take configuration backup.
- Plan downtime (if required).
- Verify drivers and firmware.
- Ensure vCenter compatibility (if managed).
- Test in lab environment first (if possible).
- After upgrade, verify hosts, networking, VMs and services.

VMWARE QUICK REVISION

Key Concepts, Architecture, ESXi, Networking and Management (Phase 1)

1 Virtualization

- Virtualization allows one physical system to run multiple virtual machines (VMs).
- Each VM has its own OS, applications and resources.
- Better hardware utilization, lower cost and easy testing.

Physical Server
(CPU, RAM, Storage, NIC)

Hypervisor

VM1	VM2	VM3
Linux	Windows	Rocky Linux

One physical machine → Multiple VMs

2 Hypervisor

- Software/platform that manages and allocates physical hardware resources to VMs.
- Divides CPU, RAM, storage and networking among multiple VMs.

Physical Hardware
(CPU | RAM | Storage | NIC)

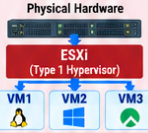
HYPERVISOR
(e.g., VMware ESXi)

VM1 VM2 VM3

3 Type 1 vs Type 2

Type 1 (Bare-Metal)

- Runs directly on physical hardware.
- No general-purpose OS required.
- Common in servers and data centers.



Type 2 (Hosted)

- Runs on top of an existing OS.
- Installs as an application.
- Common for desktops and labs.

4 VMware Introduction

- VMware has been a leading virtualization vendor and is now part of Broadcom.
- Important technologies/products:

ESXi — Bare-metal hypervisor (Type 1)

vCenter / vSphere — Centralized management

vSAN — Software-defined storage

NSX — Network virtualization and security

vmware → Now part of BROADCOM

Our current focus is VMware ESXi.

5 ESX vs ESXi

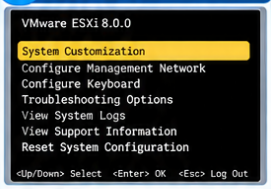
ESX (Older)	ESXi (Modern)
Older architecture.	Newer, purpose-built.
Includes Service Console.	No traditional Service Console.
Less efficient.	Smaller, more secure.
Service Console (Linux-based)	Management Components (e.g., DCUI, ESXi Shell)
VMkernel	VMkernel
Physical Hardware	Physical Hardware

Remember: ESX = old | ESXi = modern VMware bare-metal hypervisor

6 ESXi Installation

1. Boot from ESXi ISO (USB / DVD).
2. Run ESXi Installer.
3. Select target disk.
4. Install ESXi.
5. Reboot.
6. ESXi Host ready.

7 DCUI (Direct Console UI)



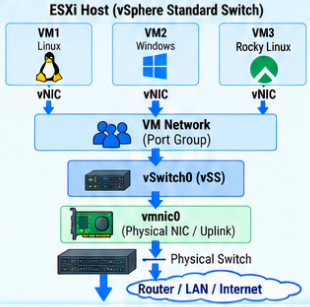
Used for:

- Management network (IP, subnet, gateway, DNS)
- Hostname
- Restart management network
- Troubleshooting options

8 Host Client vs vSphere Client

ESXi Host Client (Standalone)	vSphere Client (With vCenter)
Manage a single ESXi host.	Manage multiple ESXi hosts.
Access via web browser.	Access via vCenter.
No vCenter required.	Provides centralized management.
Administrator PC (Web Browser)	Administrator PC (Web Browser)
https://ESXi-IP	https://vCenter-IP
ESXi Host	vCenter Server
	ESXi Host 1, ESXi Host 2, ESXi Host 3

9 ESXi Networking (vSS)



10 vNIC vs vmnic

vNIC (Virtual NIC)	vmnic (Physical NIC)
Virtual machine ka virtual network adapter.	ESXi host ki physical NIC ko ESXi mein represent karne wala naam.
Exists inside the VM.	Used as uplink for vSwitch.
Connects to a VM Port Group on a vSwitch.	Connects to the physical switch.
Inside a VM: VM — vNIC0, vNIC1	**On ESXi Host:** vmnic0, vmnic1, vmnic2 ...

11 VM Port Group

- Collection of vNICs with the same network settings.
- Helps in network organization, VLAN configuration and policies.

VM1 VM2 VM3

VM Network Port Group
(e.g., VLAN 10)

vSwitch0 (vSS)

12 VMkernel Port

- Used by ESXi host for its own networking and services.
- Separate from VM Port Group.
- Used for management, vMotion, storage (iSCSI/NFS), FT, etc.

ESXi Host

vmk0
IP: 192.168.1.50

Management Network
(VMkernel Port)

vSwitch0 (vSS)

vmnic0
Physical Network)

13 NIC Teaming (Multiple Uplinks)

Single Uplink (No Redundancy)

vSwitch0 → vmnic0 → Physical Switch

If vmnic0 or link fails, connectivity will be lost. No redundancy.

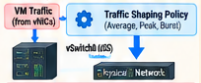
Multiple Uplinks (NIC Teaming)

vmnic0, vmnic1 → Physical Switch

- Provides redundancy / failover.
- Can distribute traffic (depending on policy).
- Better availability for production.

14 Traffic Shaping

- Controls the network traffic / bandwidth rate for a port group.
- Helps to limit or guarantee bandwidth for VMs.
- Configured at the Port Group level.



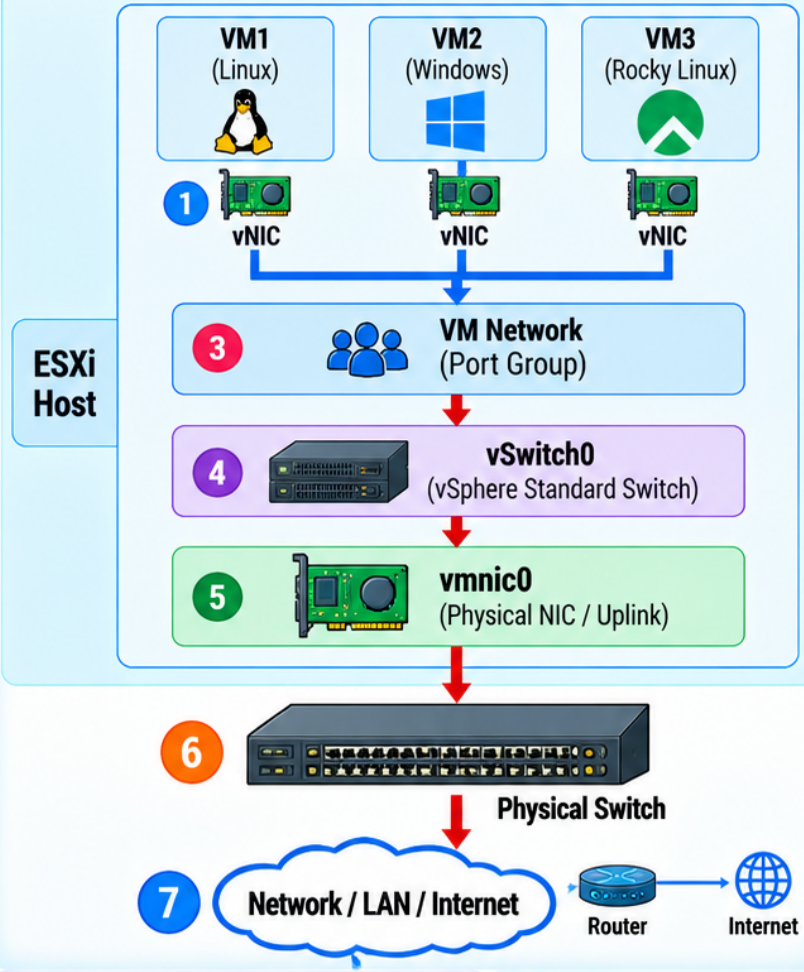
Key Parameters

Average Bandwidth	Normal/expected bandwidth (in Kbps or Mbps).
Peak Bandwidth	Maximum allowed bandwidth.
Burst Size	Temporary allowance for traffic spikes.

Question 20: VM Traffic Path to Physical Network

Step-by-Step Flow from VM to the Physical Network (Phase 1)

Inside the ESXi Host (vSphere Standard Switch)



Step-by-Step Explanation

1 — VM and vNIC

- Each VM has a vNIC (Virtual Network Interface Card).
- The vNIC is a virtual network adapter inside the VM.

2 & 3 — VM Port Group

- The VM's vNIC connects to a VM Port Group.
- Port Group provides network settings like VLAN, security and policies.
- Multiple VMs can connect to the same Port Group.

4 — vSwitch

- The VM Port Group is part of a vSphere Standard Switch (vSS).
- The vSwitch acts as a Layer 2 virtual switch inside the ESXi host.

5 — vmnic (Uplink)

- The vSwitch maps to a physical NIC (vmnic).
- vmnic is the physical network adapter/uplink on the ESXi host.

6 — Physical Switch

- The physical NIC (vmnic) connects to a physical switch.
- The switch forwards the traffic to the network.

7 — Network

- The traffic then reaches the network (LAN / router / internet) as per the network configuration.

Final Answer (Expected Flow)

VM → vNIC → VM Port Group → vSwitch → vmnic → Physical Switch → Network

Key Points

- VM uses a vNIC, not a physical NIC.
- vNIC connects to a VM Port Group.
- Port Group is part of a vSwitch.
- vSwitch maps to a vmnic (physical NIC).
- vmnic connects to the physical switch.
- Physical switch forwards traffic to the network.
- Network can be LAN, router or internet.

Real Example Flow

Command / Management View (Conceptual)

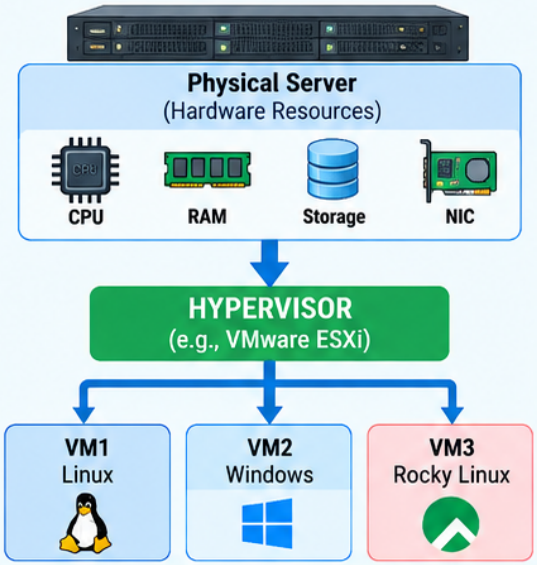
VM	VM Port Group	vSwitch	vmnic	Physical Switch	Network
(vNIC)	(e.g., VM Network)	(e.g., vSwitch0)	(e.g., vmnic0)	(Network Switch)	(LAN / Router)

VIRTUALIZATION BASICS – VISUAL GUIDE

From One Physical Machine to Multiple Virtual Machines

1 What is Virtualization?

Virtualization is a technology that allows us to use the CPU, RAM, storage and networking resources of a physical computer/server to create multiple virtual computers.
These virtual computers are called **Virtual Machines (VMs).**



One physical machine → Multiple virtual machines

2 Virtual Machine (VM)

A Virtual Machine (VM) is a software-based computer. Just like a physical computer, it has its own resources, operating system and applications. These resources are provided by the physical host through the hypervisor.

Components of a VM

- **vCPU** — Virtual CPU (processing power)
- **RAM** — Allocated memory (e.g., 4 GB)
- **Virtual Disk** — Storage (e.g., 50 GB)
- **vNIC** — Virtual Network Interface Card
- **Operating System** (e.g., Linux, Windows)
- **Applications** — Installed inside the VM

Example VM Configuration

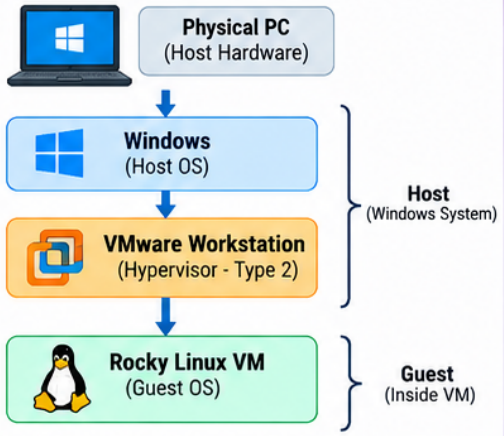
VM-01

- 2 vCPU
- 4 GB RAM
- 50 GB Virtual Disk
- 1 vNIC
- Rocky Linux (OS)

3 Host vs Guest

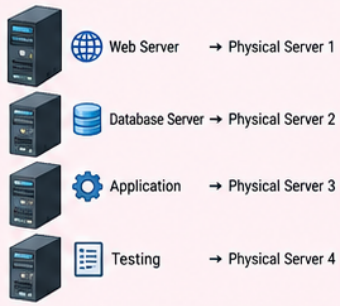
Host	Guest
• The system that provides virtualization resources. • Runs the hypervisor. • Can be a physical server (e.g., ESXi) or a desktop (e.g., Windows with Workstation).	• The operating system running inside a VM. • Uses virtual resources (vCPU, RAM, disk, vNIC). • Examples: Linux, Windows, Rocky Linux, etc.

Example: Desktop Virtualization (Type 2)

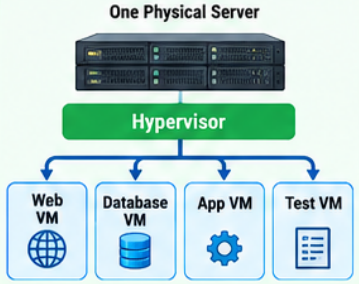


4 Benefits of Virtualization

Without Virtualization
(One server per workload)



With Virtualization
(Multiple VMs on one server)

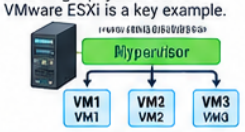


Key Benefits

- Better hardware utilization
- Multiple operating systems on one server
- Reduced hardware cost
- Easy VM create/delete
- Testing and lab environments
- VM cloning
- Snapshots
- Workload isolation
- Backup, recovery and migration options

5 Types of Virtualization

1. Server Virtualization
Multiple virtual servers on a single physical server. VMware ESXi is a key example.



2. Desktop Virtualization
Provides desktop environments centrally/virtually.

3. Network Virtualization
Network components/functions are virtualized in software.

4. Storage Virtualization
Physical storage resources are logically organized/combined.

5. Application Virtualization
Applications are delivered through virtualization instead of traditional local installation.

vSphere Standard Switch (vSS) – Complete Visual Guide

vSS Architecture, vNIC vs vmnic, Port Groups, VMkernel Port, NIC Teaming and Traffic Shaping

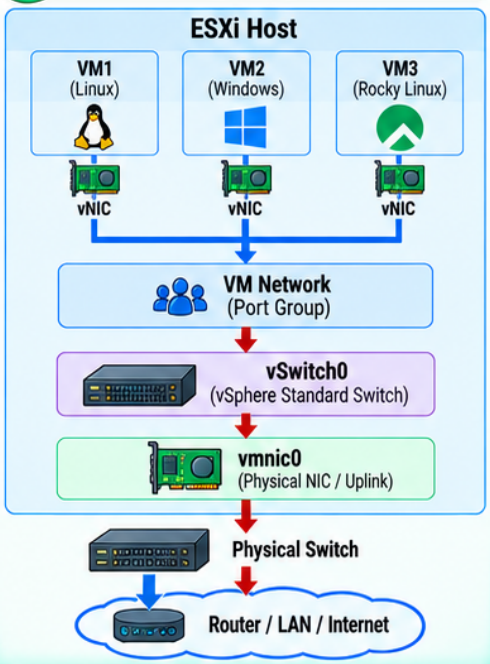
1 What is vSphere Standard Switch (vSS)?

- vSphere Standard Switch (vSS) is a software-based Layer 2 virtual switch inside an ESXi host.
- It allows VMs and the ESXi host (via VMkernel ports) to connect to the physical network using vmnic uplinks.
- vSS runs inside the ESXi host and provides basic virtual networking features like port groups, VLANs and policies.

Key Points

- Software-based Layer 2 switch inside ESXi.
- Connects VMs (vNICs) and VMkernel ports.
- Uses vmnic (physical NIC) for uplink.
- Provides port groups for network organization.
- Supports VLANs and basic network policies.

2 vSS Networking Architecture



3 vNIC vs vmnic

vNIC (Virtual NIC)

- Virtual Machine ka virtual network adapter.
- Exists inside the VM.
- Connects to a VM Port Group on a vSwitch.
- Each VM can have one or more vNICs.

Example inside a VM

VM — vNIC0, vNIC1

vmnic (Physical NIC)

- ESXi host ki physical NIC ko ESXi mein represent karne wala naam.
- Used as uplink for vSwitch.
- Connects to the physical switch.
- Provided by the physical hardware.

Example on ESXi Host

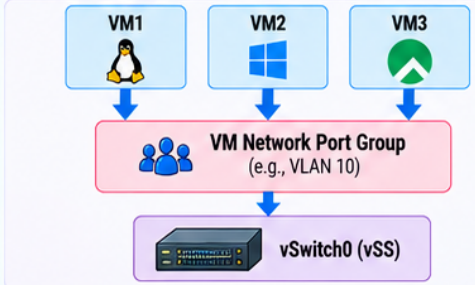
vmnic0
vmnic1
vmnic2 ...

Quick Flow: VM to Physical Network

VM (vNIC) → Port Group (e.g., VM Network) → vSwitch0 (vSS) → vmnic0 (Physical NIC) → Physical Switch

4 VM Port Group

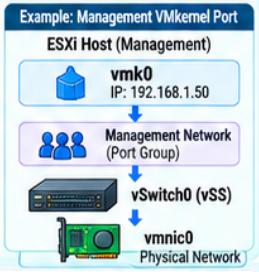
- A VM Port Group is a collection of vNICs with the same network settings.
- VMs normally connect to a VM Port Group on a vSwitch.
- Port Groups help in network organization, VLAN configuration and network policies (e.g., security, traffic shaping).



5 VMkernel Port

- VMkernel Port is used by the ESXi host for its own networking and services.
- Used for management traffic, vMotion, storage (iSCSI/NFS), FT, etc.
- Different from VM Port Group (used by VMs).
- Each VMkernel port has a vmk interface (e.g., vmk0, vmk1).

Example: Management VMkernel Port



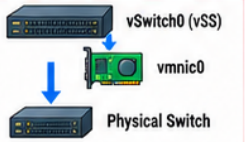
6 VM Port Group vs VMkernel Port

	VM Port Group (For VMs)	VMkernel Port (For ESXi Host)
Used by	Virtual Machines (vNICs)	ESXi Host (services)
Purpose	VM network traffic (e.g., production, test)	Management, vMotion, Storage (iSCSI/NFS), FT, etc.
Example	VM Network, App Network	Management Network, vMotion Network
Interface Name	N/A (used by vNIC)	vmk0, vmk1, vmk2 ...
Configuration	VLAN, security, traffic shaping, etc.	IP address, VLAN, services
Traffic Type	Guest VM traffic	ESXi host/service traffic

Golden Rule:
VM ko network chahiye → VM PORT GROUP
ESXi host/service ko network chahiye → VMKERNEL PORT

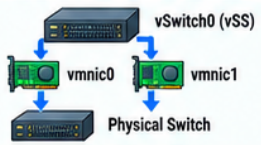
7 NIC Teaming (Multiple Uplinks)

Single Uplink (No Redundancy)



- If vmnic0 or link fails, connectivity will be lost.
- No redundancy.

Multiple Uplinks (NIC Teaming)

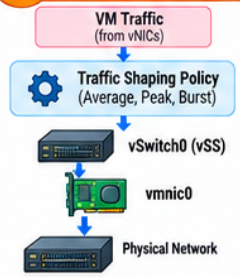


- Provides redundancy / failover.
- Can distribute traffic (depending on teaming policy).
- Better availability for production.

8 NIC Teaming Policies (Common)

Policy	Description
Load Balancing	How outgoing traffic is distributed (e.g., based on port ID, MAC, IP).
Network Failover Detection	Detects link or device failure.
Notify Switches	Yes/No (affects how switches handle NIC teaming).
Failback	Whether to automatically use a link again after failure.
Active Adapters	Uplinks used for active traffic.
Standby Adapters	Uplinks kept as standby (for failover).

9 Traffic Shaping (Port Group Level)



- Traffic shaping controls the network traffic/bandwidth rate for a port group.
- Helps to limit or guarantee bandwidth for VMs.
- Can be configured at the Port Group level.

Key Parameters

Average Bandwidth	Normal/expected bandwidth (in Kbps or Mbps).
Peak Bandwidth	Maximum allowed bandwidth.
Burst Size	Temporary allowance for traffic spikes.